Water Research 308 (2027) 126838

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

Micro-aeration unlocks endogenous nitrogen removal in anaerobic digesters via metabolically versatile fermentative bacteria

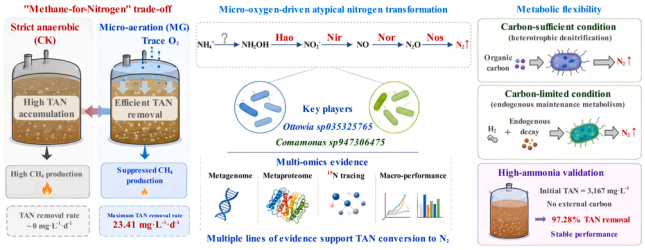
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HIGHLIGHTS

- ^{15}N tracing confirmed ammonium-derived N_2 formation under micro-aeration.
- Fermentative taxa were associated with coupled carbon and nitrogen metabolism.
- Carbon limitation induced endogenous resource-mobilization responses.
- Micro-aeration caused a clear methane-nitrogen trade-off.
- A side-stream configuration is proposed to separate N removal from methanogenesis.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Ammonia inhibition
Anaerobic digestion
Micro-aeration
Meta-omics analyses
Metabolic flexibility
Nitrogen removal

ABSTRACT

Ammonia inhibition represents a significant bottleneck in the anaerobic digestion (AD) of high-nitrogen organic wastes. This study employed a batch reactor system subjected to alternating carbon-sufficient and carbon-limited phases to investigate whether pulsed micro-aeration, with a peak oxidation–reduction potential ranging from −150 to −50 mV, could promote the conversion of total ammonia nitrogen (TAN) to N_2 in anaerobic digesters maintained under bulk anaerobic conditions, while simultaneously elucidating the competitive interplay between this nitrogen removal process and methanogenesis. The investigation integrated reactor performance monitoring, ^{15}N stable-isotope tracing, and combined metagenomic and metaproteomic analyses. Compared to the strictly anaerobic control, the pulsed micro-aeration group exhibited a significant decrease in TAN concentration, with a maximum removal rate of 23.41 $mg{\cdot}L^{-1}{\cdot}d^{-1}$. Isotope analysis revealed that approximately 82.49% of the transformed $^{15}NH_4^+$-N was recovered as $^{15}N_2$, thereby confirming the conversion of TAN to N_2. The transient accumulation of NO_2^- and NH_2OH, coupled with the absence of sustained NO_3^- accumulation and the enhanced expression of Hao, Nir, Nor, and Nos proteins, collectively suggested the involvement of a hydroxylamine/nitrite-associated pathway in conjunction with denitrification within the micro-aerated system. This nitrogen removal process was primarily mediated by native fermentative/facultative heterotrophic bacteria in the AD system, including *Ottowia* and *Comamonas*, which exhibited pronounced metabolic flexibility in response to carbon availability. When carbon was abundant, nitrogen removal was predominantly driven by heterotrophic denitrification; under carbon limitation, the system transitioned towards endogenous maintenance modes characterized by potential hydrogen utilization, mobilization of intracellular reserves (such as polyhydroxybutyrate and fatty acids), scavenging of residual organic matter, and the glyoxylate cycle. However, the

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https://doi.org/10.1016/j.watres.2026.126838
Received 25 June 2026; Received in revised form 24 August 2026; Accepted 30 August 2026
Available online 31 August 2026
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activation of nitrogen removal through micro-aeration was accompanied by a suppression of methanogenesis, resulting in a methane loss of 1.31–2.77 mL per mg of TAN removed. Consequently, a paradigm of "mainstream methanogenesis–side-stream micro-aerobic nitrogen removal" is proposed, offering a novel strategy for developing robust processes to mitigate ammonia inhibition in AD.

1. Introduction

Anaerobic digestion (AD) is a fundamental technology for the recycling of organic solid waste and has gained increasing prominence under China's dual-carbon strategy (Hoy et al., 2023; Xing et al., 2022). However, ammonia inhibition has emerged as a critical bottleneck limiting the stable operation of AD when treating nitrogen-rich organic wastes. Biological nitrogen removal technologies—including nitrification-denitrification, partial nitrification-denitrification, anammox, and ferric ammonium oxidation (Feammox)—have been widely applied in landfill leachate and municipal wastewater treatment due to their efficiency and cost-effectiveness (Xiong et al., 2024; Zhao et al., 2024). Nevertheless, the reaction conditions required by these technologies fundamentally conflict with the environmental characteristics of AD reactors. Specifically: (1) Nitrification necessitates sufficient dissolved oxygen, whereas AD operates under strict anaerobic conditions (Kuypers et al., 2018); (2) Denitrification and anammox depend on nitrate or nitrite as electron acceptors, yet AD systems lack sustained pathways to supply these oxidized nitrogen species (Su et al., 2024); (3) Denitrification usually requires organic carbon as an electron donor, which contradicts the core objective of AD—directing organic carbon toward methane generation (Peng et al., 2025; Wang et al., 2025). Furthermore, high-concentration organic carbon environments suppress autotrophic anammox and Feammox-associated microbes, thereby restricting the feasibility of these autotrophic nitrogen removal pathways in AD reactors (Cui et al., 2025; Xia et al., 2023; Zhao et al., 2024). Given these limitations, sustained *in situ* biological removal of total ammonia nitrogen (TAN) is generally constrained in conventional AD systems, where TAN tends to accumulate. Achieving *in situ* and efficient TAN removal while maintaining methane recovery remains an unresolved engineering and scientific challenge in this field.

In recent years, studies investigating the intensification of AD process have unexpectedly revealed that micro-aeration, while facilitating hydrolysis and acidogenesis, is associated with a significant reduction in TAN levels. For instance, Liu et al. (2024) reported that the maximum TAN concentration in the micro-aerated group was only 75.60% of that observed in the control group, a discrepancy that cannot be fully explained by ammonia stripping alone. Furthermore, low-dose oxygen input has been shown to alter the redox state, hydrolysis and acidogenesis processes, and nitrogen speciation within AD systems (Liang et al., 2025). Consistent with these findings, we previously observed that in our long-term operated food waste AD reactors, the TAN concentration, while generally accumulating, was intermittently punctuated by abrupt declines of approximately 28.94%–33.33% (Peng et al., 2018; Zhang et al., 2022). These declines coincided temporally with unintentional oxygen ingress or occasional aeration events. However, these observations remain operational phenomena or correlative evidence. To elucidate the underlying mechanisms, this study proposes a hypothesis: low-dose oxygen input may induce transient redox perturbations within the anaerobic environment, thereby creating a reaction window for ammonia transformation. Nevertheless, whether ammonia is ultimately converted to N_2 under micro-aeration, which microorganisms are involved in this process, how their functional activities respond to micro-aeration and carbon availability, and whether a trade-off exists between this process and methanogenesis all require systematic validation.

To address these questions systematically, this study employed a batch reactor system subjected to alternating carbon-sufficient and carbon-limited phases to investigate whether pulsed micro-aeration could promote the conversion of TAN to N_2 in anaerobic digesters and elucidate the trade-off between this process and methanogenesis. This study integrated reactor performance monitoring, ^{15}N stable-isotope tracing, computational fluid dynamics (CFD)-assisted redox profiling, extracellular polymeric substances (EPS) characterization, and combined metagenomic and metaproteomic analyses. Initially, a multiphase, well-controlled model system was employed to elucidate the fundamental patterns and microbial mechanisms associated with micro-aeration-induced nitrogen removal. Subsequently, an independent validation experiment was conducted under high-ammonia conditions (3167.44 $mg{\cdot}L^{-1}$) to assess the applicability of this process in realistic inhibitory environments. This research aimed to provide a comprehensive understanding of micro-oxygen-driven *in situ* nitrogen removal in AD, thereby establishing a scientific foundation for the development of next-generation AD processes that can mitigate ammonia inhibition.

2. Materials and methods

2.1. Feedstock, inoculum, and isotope tracer

Synthetic food waste was used as the feedstock and was prepared according to a method established based on the characteristics of actual food waste (Ye et al., 2024). The feedstock was composed of a mixture of vegetables, meat, cooked rice, and fruit peels in wet-weight ratios of 40%, 15%, 25%, and 20%, respectively. The elemental composition, carbon-to-nitrogen (C/N) ratio, and other physicochemical properties of the synthetic food waste are provided in Supplementary Table 2.1. The inoculum consisted of anaerobic sludge collected from a municipal wastewater treatment plant, which was acclimated in a laboratory-scale anaerobic digester at 37 °C with a loading rate of 1.5 g VS $(L{\cdot}d)^{-1}$ for 30 days until both methane production and total nitrogen content stabilized. The stable-isotope tracer used was ^{15}N-labeled ammonium chloride ($^{15}NH_4Cl$, ^{15}N abundance $\geq$ 98%).

2.2. Experimental setup and microaeration strategy

The experiments were conducted using 500 mL serum bottles with a working volume of 450 mL, maintained at a constant temperature of 37 °C. Each reactor was equipped with a fine-pore diffuser, a liquid sampling port, and a gas collection outlet (Fig. 1a). All reactors were sparged with argon to establish an initial anaerobic atmosphere. Three sets of reactors were established: a blank control (CK), a micro-aeration test group (MG), and a heat-inactivated control (SMG), in which the sludge was autoclaved at 121 °C for 30 min (Wang et al., 2023). The SMG operated under the same micro-aeration conditions as the MG, where heat inactivation was applied to suppress the activity of methanogens and most metabolically active microorganisms, thereby providing a reference for estimating ammonia stripping and other physicochemical TAN losses. Micro-aeration was achieved by daily injections of pure oxygen ($\geq$ 99.995%) via a syringe needle.

The experiment was operated in batch mode with distinct phases, during which the reactor contents were retained throughout the entire evaluation period without intermediate feeding or discharge, except for periodic sampling. Only the intensity of the oxygen supply or carbon availability was adjusted between phases. Specifically, the experiment consisted of four phases (Fig. 1b). In Phase I (days 0–12), oxygen was supplied at a rate of 0.095 L $O_2{\cdot}(L{\cdot}d)^{-1}$ to maintain a daily oxidation–reduction potential (ORP) peak of approximately –150 mV, with the objective of determining whether simultaneous ammonia removal

and methane production could be achieved. In Phase II (enrichment phase, days 12–32), the oxygen supply was increased to 0.190 L O_2·(L·d)$^{-1}$ to elevate the daily ORP peak to about –50 mV. This increase in oxidative pressure aimed to further select and enrich nitrogen-removing bacteria adapted to micro-aeration conditions and to verify the stability of the newly established function under intensified aeration. Phases III and IV (flexibility-testing phases, days 33–40) were designed to evaluate the metabolic flexibility and reversibility of the system. In each of these phases, 0.5 g of glucose—rather than food waste—was added once at the start, under the same aeration conditions as in Phases I and II, respectively. Glucose was chosen over food waste because it is a chemically defined, nitrogen-free, and readily biodegradable carbon source, thereby avoiding the reintroduction of nitrogen and allowing the assessment of the system's response solely to changes in carbon availability. Although the ORP peaks generated by pulsed aeration varied among operational phases, the reactors were not operated with continuous oxygen supply to fully oxidize ammonia at any stage. Instead, transient redox perturbations were imposed against an overall anaerobic background to provide reaction windows for ammonia transformation.

2.3. Stable isotopic labeling test

On day 40, the sludge from the MG group was anaerobically washed to eliminate residual nitrogen and was subsequently transferred to a new reactor containing synthetic wastewater, as detailed in Supplementary Method 1.1. Micro-aeration was applied, and the transformation pathway of $^{15}NH_4^+$ in the reactor was subsequently tracked (Fig. 1c). The analytical methods and sampling protocols are outlined in Supplementary Method 1.2.

2.4. Assessment of process robustness and nitrogen removal boundaries under high-ammonia stress

Following the phased batch and isotope-tracing experiments, the feasibility of the micro-aeration process for treating actual digestate was evaluated. A 50-day batch experiment was conducted using real digestate sourced from a full-scale AD plant in Luzhou, Sichuan, China, with an initial chemical oxygen demand (COD) of 4193.70 mg·L^{-1} and TAN of 3167.44 mg·L^{-1}. The detailed experimental design is presented in Supplementary Method 1.3.

2.5. Physicochemical analysis

The ORP was continuously monitored using an online electrode positioned 2 cm below the liquid surface, with data recorded at 10 s intervals. Liquid-phase samples were collected periodically, while gas-phase samples were collected at the end of each experimental stage. The liquid samples were analyzed for COD, TAN, total nitrogen, nitrate nitrogen, nitrite nitrogen, and hydroxylamine, whereas the gas samples were analyzed for methane and hydrogen. Mixed sludge samples were collected at the end of Phases II, III, and IV for EPS extraction, scanning electron microscopy (SEM) and confocal laser scanning microscopy (CLSM) analyses. EPS was sequentially fractionated into soluble EPS (S-EPS), loosely bound EPS (LB-EPS), and tightly bound EPS (TB-EPS) according to Ye et al. (2025). Briefly, mixed sludge samples were centrifuged, and the resulting supernatant was collected as S-EPS. The pellet was resuspended in preheated NaCl solution, vortex-mixed, and centrifuged to obtain LB-EPS. The remaining pellet was subsequently incubated in a 60 °C water bath and centrifuged to obtain TB-EPS. All EPS extracts were filtered through 0.45 μm membrane filters. Polysaccharides, proteins, humic substances, and DNA were quantified using the phenol-sulfuric acid method (DuBois et al., 1956), Lowry-Folin method (Lowry et al., 1951), modified Lowry method (Frølund et al., 1996), and diphenylamine method (Burton, 1956), respectively. For SEM analysis, samples were fixed with glutaraldehyde, post-fixed with osmium tetroxide, dehydrated through a graded ethanol series, subjected to critical-point drying, and sputter-coated with gold before observation using a HITACHI SU8010 scanning electron microscope. For CLSM analysis, SYTO 63, FITC, iFluor 555-ConA, and Calcofluor White were used to label cells, proteins, α-polysaccharides, and β-polysaccharides, respectively. Images for all groups were acquired under identical imaging parameters. Detailed analytical procedures for the individual parameters are provided in Supplementary Method 1.4.

2.6. Metagenomic and metaproteomic analyses

Sludge samples intended for metagenomic sequencing, metagenome-assembled genome (MAG) reconstruction, and metaproteomic analysis were collected at the conclusion of Phases II and IV. These two phases represent the functional response characteristics induced by micro-aeration under carbon-limited and carbon-sufficient conditions, respectively. Comparing these two phases facilitates the differentiation of micro-aeration-induced nitrogen transformation characteristics under carbon-limited versus carbon-sufficient conditions, thereby elucidating

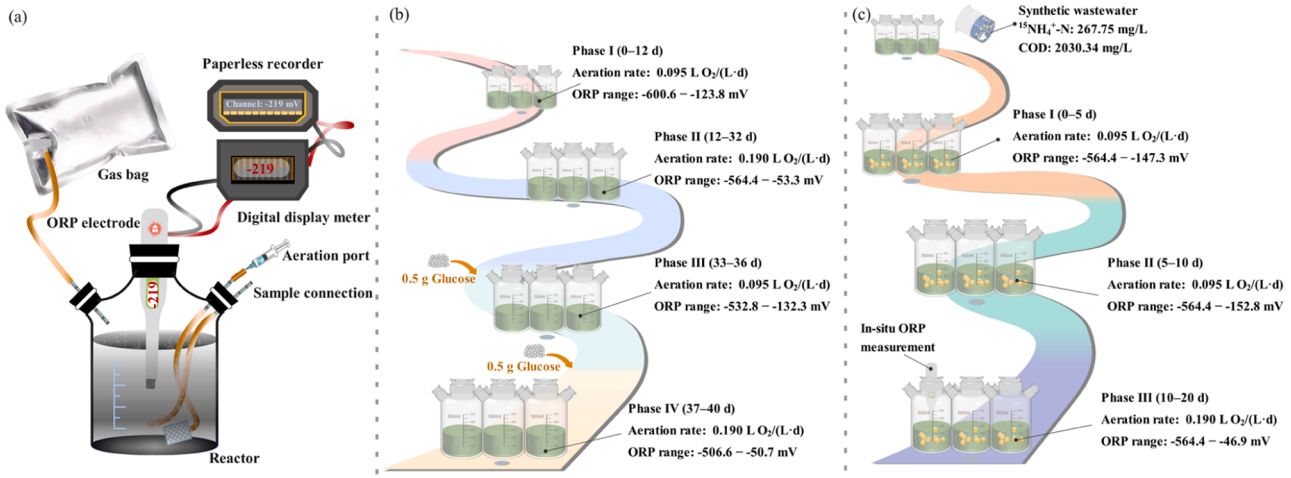


Fig. 1. Experimental setup and workflows of the batch and ^{15}N isotope-tracing experiments. (a) Schematic diagram of the reactor configuration and real-time ORP monitoring system. (b) Four-phase batch experiment showing the phase-specific aeration rates, glucose addition events, and observed ORP ranges. (c) Three-phase ^{15}N isotope-tracing experiment showing the synthetic wastewater addition, aeration conditions, and observed ORP ranges. Observed ORP ranges indicate the minimum to maximum values recorded during each phase.

the functional linkages between carbon metabolism and nitrogen transformation. Detailed procedures for DNA sequencing, MAG binning, and phylogenetic assignment are outlined in Supplementary Method 1.5, while metaproteomic analysis procedures are described in Supplementary Method 1.6.

2.7. Data processing

CFD simulations were conducted to assess oxygen mass transfer (Supplementary Method 1.7). Both first- and second-order kinetic models were employed to analyze the ammonia removal process (Supplementary Method 1.8). The calculations for the ^{15}N mass balance were based on the ^{15}N atom percent abundance and the concentrations of individual nitrogen pools (Supplementary Method 1.9). Density functional theory calculations were executed using ORCA software (version 6.0.0) (Neese, 2025), with model molecules selected based on existing literature (Zhang et al., 2025). The apparent methane–nitrogen trade-off coefficient (Θ), defined as the volume of methane lost per unit mass of TAN removed (mL CH_4 mg^{-1} TAN), was calculated to quantify the trade-off between nitrogen removal and methanogenesis. The calculation procedure is detailed in Supplementary Method 1.10. All statistical analyses, including mean ± standard deviation, as well as data visualizations, were performed using RStudio, Origin 2024b, and Python.

3. Results

3.1. Micro-aeration induces a "Methane-for-nitrogen" functional shift

The introduction of trace oxygen successfully initiated the biological conversion of ammonia while simultaneously causing significant inhibition of methanogenesis, thereby establishing the fundamental trade-off between methane production and nitrogen removal (Figs. 2a and 2b).

During the substrate-sufficient start-up phase (Phase I), the concentration of TAN in the MG group was significantly lower than that in the CK group ($p < 0.001$, Fig. 2a, Supplementary Table 2.2). This indicates that the nitrogen removal process was activated immediately following the initiation of micro-aeration. As operations transitioned into the carbon-limited endogenous metabolism phase (Phase II, Fig. 2c), the nitrogen removal rate in the MG group reached 20.07 $mg{\cdot}L^{-1}{\cdot}d^{-1}$, accompanied by the accumulation of endogenous hydrogen (Fig. 2b). The decline in TAN during this phase exhibited second-order empirical kinetics ($R^2 = 0.97$; Fig. 2d, Supplementary Table 2.3), suggesting that the process may be co-limited by TAN concentration and the availability

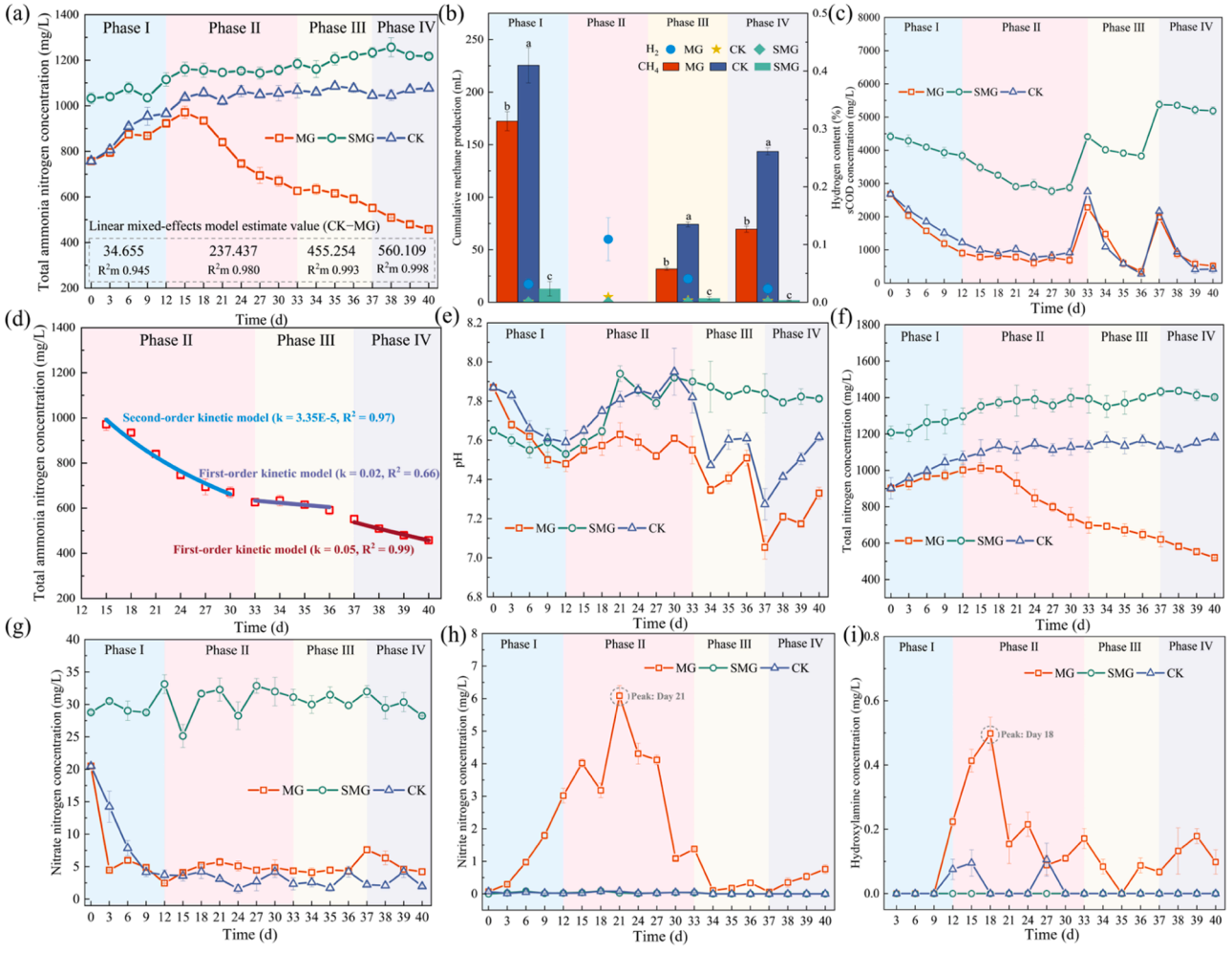


Fig. 2. Nitrogen removal and methanogenic performance under stepwise micro-aeration. (a) Temporal profiles of TAN concentrations across the four operational phases. Phase-specific CK–MG estimates and marginal coefficients of determination (R^2m) from linear mixed-effects models are shown at the bottom ($p < 0.001$; Supplementary Table 2.2). (b) Cumulative methane production (bars) and H_2 content (symbols) in the MG, CK, and SMG groups. Different lowercase letters indicate significant differences among groups ($p < 0.05$). (c) Temporal profiles of COD concentrations. (d) Phase-specific kinetic fitting results of TAN removal in the MG group, with a second-order model for Phase II and first-order models for Phases III and IV (k, rate constant; R^2, coefficient of determination). (e) Temporal variation in pH. (f) Temporal profiles of total nitrogen (TN). (g–i) Temporal profiles of nitrate (g), nitrite (h), and hydroxylamine (i), with the peak accumulation times of nitrite and hydroxylamine indicated, respectively.

of reducing equivalents. Upon the addition of glucose in Phases III and IV, the system exhibited a rapid response, as evidenced by a sharp decline in pH (Fig. 2e), while maintaining its nitrogen removal capability, albeit with varying rates: decreasing to 8.81 $mg{\cdot}L^{-1}{\cdot}d^{-1}$ in Phase III, yet surging by 165.72% to 23.41 $mg{\cdot}L^{-1}{\cdot}d^{-1}$ in Phase IV when aeration intensity was elevated (Fig. 2d). This indicates that removal activity is positively regulated by oxygen availability. The removal rates in both phases adhered to first-order kinetics (R^2 = 0.66 and 0.99, Supplementary Table 2.3), likely due to the alleviation of electron donor limitations compared to Phase II.

Monitoring of nitrogen intermediates provided valuable insights into the nitrogen transformation process. The trend of total nitrogen in the liquid phase closely mirrored that of TAN (Fig. 2f), and no sustained accumulation of nitrate was observed (Fig. 2g). However, transient appearances of nitrite and hydroxylamine at low concentrations were detected (Figs. 2h and 2i), suggesting the transient formation of nitrogen transformation intermediates without sustained accumulation of oxidized nitrogen species. The ^{15}N stable isotope tracing experiment further elucidated the ultimate fate of TAN (Fig. 3, Supplementary Table 2.4). Over the 20-day incubation period, 67.15% of the labeled $^{15}NH_4^+$ was transformed, with approximately 82.49% recovered as $^{15}N_2$. Additionally, around 11.22% was attributed to unaccounted nitrogen losses, which may include trace intermediates and measurement errors, while approximately 3.03% accumulated as nitrate, nitrite, and assimilated organic nitrogen. These results further confirm that micro-aeration enhances the conversion of TAN to N_2 in AD systems, with N_2 emerging as the predominant terminal product of this process.

In stark contrast to the activation of nitrogen removal, methanogenesis in the MG group was significantly suppressed. H_2 levels were notably higher in the MG group compared to the CK group across all phases (Fig. 2b). This accumulation of H_2 may be attributed to the inhibition of hydrogenotrophic methanogen activity induced by micro-aeration. Furthermore, the cumulative methane production in the MG group was consistently and significantly lower than that in the CK group throughout the experiment, with reductions of 23.54% in Phase I and 51.52%–57.11% in Phases III and IV ($p < 0.05$). This observation suggests a competitive distribution of carbon flow and reducing equivalents between the nitrogen removal and methanogenesis pathways. To further quantify this trade-off, we calculated the phase-specific apparent methane–nitrogen trade-off coefficient (Θ). The Θ values for Phases I, III, and IV were 2.77, 2.10, and 1.31 mL CH_4 mg^{-1} TAN, respectively (Supplementary Table 2.5). These results indicate that micro-aeration-enhanced TAN removal was associated with a quantifiable methane cost, with the methane loss per unit of additional TAN removed decreasing progressively across the operational phases. Importantly, to evaluate the practical potential of this process under realistic high-ammonia conditions, an independent validation experiment was conducted using actual digestate with an initial TAN concentration of 3167.44 mg L^{-1}. Without external carbon supplementation, the system achieved a TAN removal efficiency of 97.28% over a 50-day micro-aerobic operation, demonstrating its potential to mitigate ammonia inhibition (Supplementary Figure 1).

3.2. *Pulsed micro-aeration creates redox heterogeneity, restructures sludge aggregates, and drives microbial community succession*

Having established that micro-aeration triggers a "methane-for-nitrogen" functional shift, we next investigated the potential physicochemical drivers and microbial responses underlying this phenomenon. Our results indicate that intermittent micro-aeration creates dynamic redox heterogeneity within the reactor, restructures the characteristics of sludge aggregates, and drives microbial community succession centered on versatile bacteria capable of coupled carbon and nitrogen metabolism.

Low-intensity aeration was employed to introduce trace amounts of oxygen into the reactors. Online monitoring revealed that aeration induced transient upward pulses in ORP near the liquid surface, with peak values reaching approximately −150 to −50 mV, followed by a rapid decline during inter-aeration periods (Fig. 4a). In situ profile measurements demonstrated that during aeration, a continuous vertical ORP gradient developed from the top to the bottom of the reactor, with a maximum difference of 424 mV (ranging from −66 mV to −490 mV; Fig. 4b). This spatial gradient arose from limited oxygen mass transfer due to the high viscosity of the sludge (82 cP), resulting in a decreasing concentration of dissolved-oxygen with depth. CFD simulations showed that the region directly above the aerator exhibited the highest liquid-

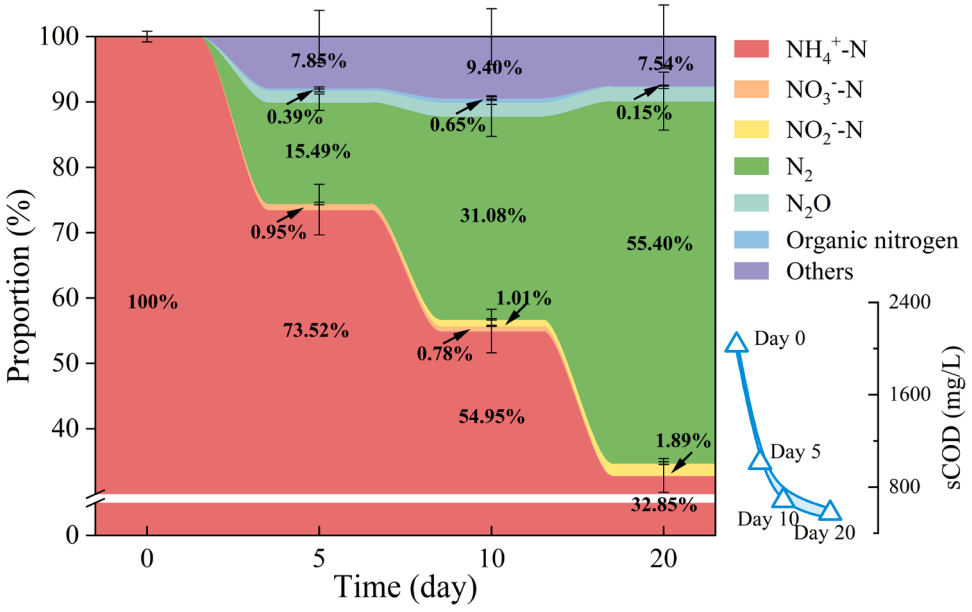


Fig. 3. Quantitative mapping of nitrogen transformation and fate using ^{15}N stable-isotope tracing. Stacked areas show the temporal evolution of the ^{15}N mass balance over the 20-day incubation. Colored segments represent the proportions of each ^{15}N-labeled nitrogen species. "Others" encompasses unmeasured intermediates, such as hydroxylamine and hydrazine, and other inorganic nitrogen fractions. The inset shows the concurrent COD degradation profile. Values in the plots indicate the mean percentage of each component at the indicated time points: days 5, 10, and 20.

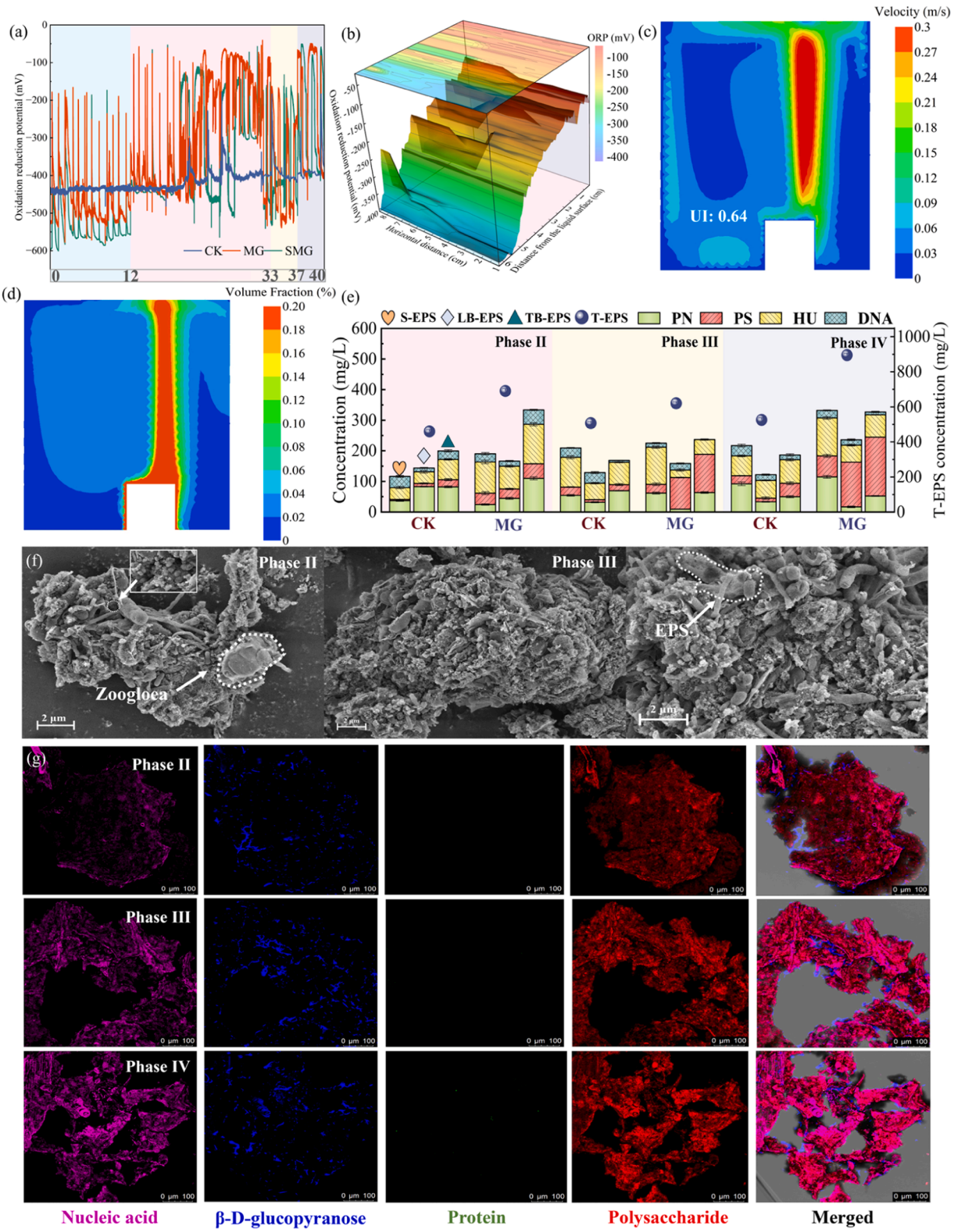


Fig. 4. Redox heterogeneity, EPS composition, and sludge aggregate structure under pulsed micro-aeration. (a) Real-time ORP monitoring throughout reactor operation. Cyclic ORP profiles in the MG group reflect dynamic redox fluctuations under pulsed micro-aeration. (b) Three-dimensional *in situ* mapping of ORP distribution, revealing vertical and horizontal ORP differences. (c, d) CFD simulations of reactor hydrodynamics, showing the axial cross-sectional distribution of the liquid-phase velocity field (c) and gas-volume fraction (d). UI denotes the velocity uniformity index calculated from the liquid-phase velocity field (0.64). (e) Composition of different EPS fractions at the end of Phases II–IV. For each group, the three bars represent S-EPS, LB-EPS, and TB-EPS. Stacked colors indicate the contents of proteins, polysaccharides, humic substances, and DNA. Dots indicate the total EPS concentration. (f) SEM images showing phase-dependent changes in sludge aggregate morphology. Dashed circles indicate representative aggregate structures. (g) CLSM images of sludge aggregates. Samples were stained for nucleic acids, β-D-glucopyranose, proteins, and polysaccharides. The merged images show the distribution of nucleic acid- and EPS-related fluorescence signals within the sludge aggregates.

phase velocity and gas holdup, which gradually decreased in both radial and axial directions (Figs. 4c and 4d). This non-uniform mixing regime (velocity uniformity index = 0.64; Fig. 4c) further indicated that significant local redox differences may exist within the reactor under micro-aeration conditions, potentially creating microenvironments with varying availabilities of electron acceptors, thereby providing distinct local conditions for microorganisms with different metabolic traits.

Under the aforementioned redox heterogeneity, phase-dependent changes were observed in the concentrations of various EPS fractions and components, as well as in the structure of sludge aggregates. By the end of Phase II, the concentration of TB-EPS in the MG group reached 333.67 mg/L, accounting for 48.32% of the total EPS (TEPS), which was significantly higher than that in the CK group (199.60 mg/L), corresponding to a 67.17% increase ($p < 0.05$; Fig. 4e and Supplementary Table 2.6). Chemical composition analysis revealed that the core components of TB-EPS were extracellular proteins (PN, 32.94%) and humic substances (HU, 38.75%) (Fig. 4e). SEM images further illustrated differences in aggregate morphology between the ends of Phases II and IV, with the MG group displaying a denser structure and more extensive EPS matrix coverage in Phase II (Fig. 4f). CLSM observations revealed phase-dependent differences in fluorescence signals corresponding to nucleic acids, β-D-glucopyranose, proteins, and polysaccharides (Fig. 4g). Notably, the fluorescence signal of β-D-glucopyranose was weakest in Phase II (Fig. 4g), consistent with the lower proportion of extracellular polysaccharides determined by chemical analysis. Following the addition of glucose in Phase III, the TB-EPS concentration in the MG group decreased significantly by 28.96% compared to Phase II (from 333.67 to 237.04 mg/L, $p < 0.05$), while the concentration of S-EPS increased significantly by 18.10% (from 190.33 to 224.78 mg/L, $p < 0.05$). By Phase IV, under intensified micro-aeration, the concentrations of S-EPS and TB-EPS converged (331.99 vs. 326.99 mg/L, $p > 0.05$; Fig. 4e), while the concentration of LB-EPS increased by 49.01% compared to Phase III. Given that TB-EPS and its protein and humic components are generally considered to be associated with sludge aggregate stability, matrix adhesion, and the maintenance of the local microenvironment (Wang et al., 2024; Wang et al., 2023), the observed restructuring of the EPS matrix may provide a structural basis for maintaining sludge aggregate integrity under micro-aeration and for microbial adaptation to redox fluctuations (Supplementary Figure 2).

To identify the microbial drivers responsible for the functional shift discussed, metagenomic sequencing and assembly were conducted on samples from Phase II (carbon-limited) and Phase IV (carbon-sufficient), resulting in the identification of 93 high-quality metagenome-assembled genomes (MAGs) (Fig. 5). A systematic comparison of the overall

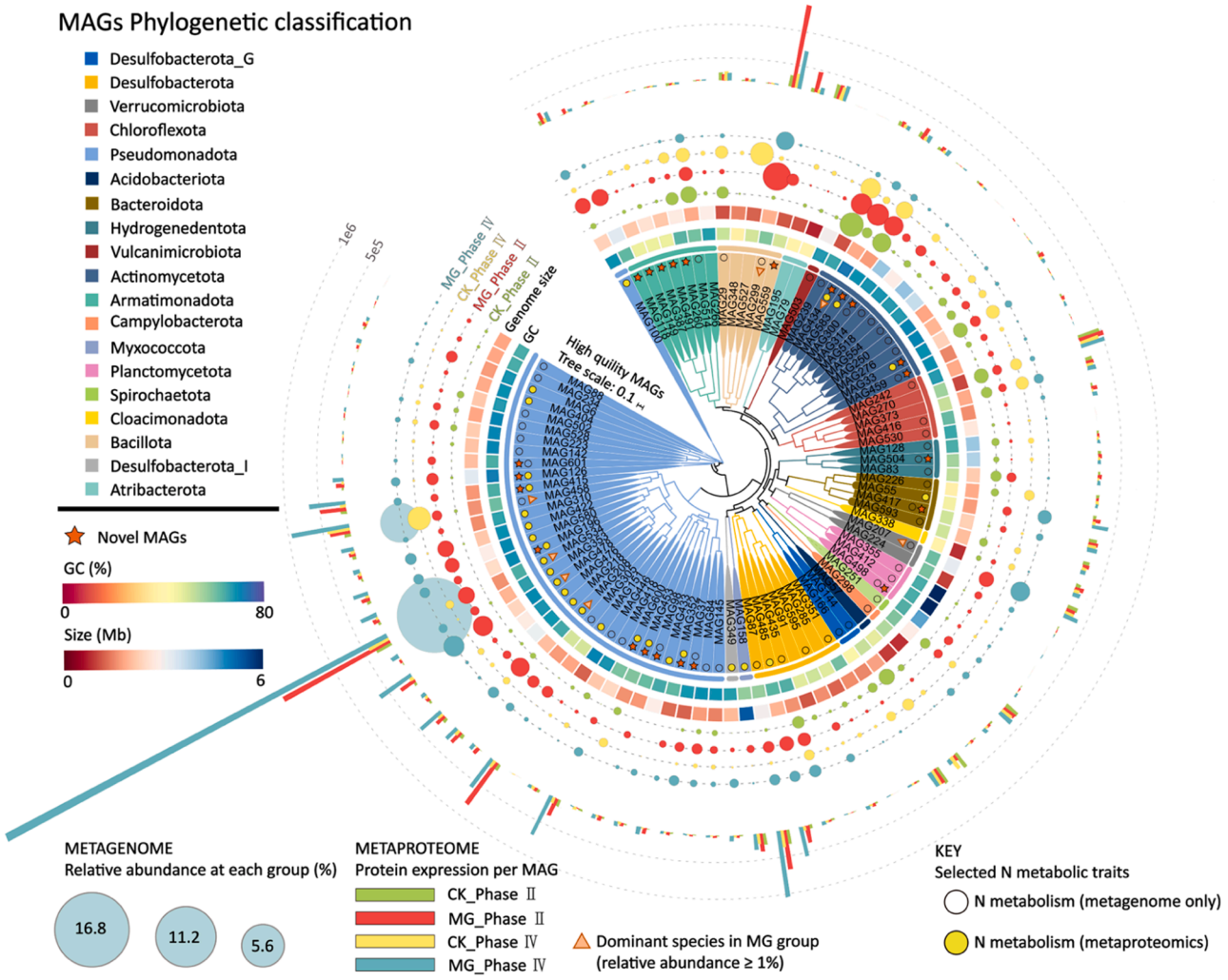


Fig. 5. MAG-centric metaproteomic analysis of the enrichment cultures. Maximum-likelihood phylogenomic tree of the MAGs, constructed from concatenated conserved protein sequences. MAGs were taxonomically classified to the phylum level using the Genome Taxonomy Database (GTDB), as indicated by branch colors and labels. Branch decorations highlight novel unclassified MAGs and those encoding nitrogen-metabolism functions. Outer heatmaps display genomic GC content and genome size. The relative abundance of each MAG, based on sequencing coverage, is shown by bubbles at the branch tips; the expression level of associated proteins, quantified by peptide peak areas, is represented by bars.

community structure between the MG and CK groups revealed that micro-aeration induced a significant succession of the microbial community. In the MG group, 46 MAGs were specifically enriched. Among these, the most prominent taxa—dominant across multiple phases (relative abundance > 1%)—included *Ottowia* sp035325765 (MAG478), *Comamonas* sp947306475 (MAG534), *Propioniciclava* sp. (MAG434), and *Yanghanlia* sp. (MAG208) (Fig. 5). A key finding is that these dominant taxa, which were preferentially selected under micro-aeration conditions, were not the typical ammonia-oxidizing or dedicated denitrifying bacteria commonly found in wastewater treatment systems. Instead, most of them are recognized as hydrolytic-acidogenic or facultative heterotrophic bacteria prevalent in AD systems (Feng et al., 2024; Li et al., 2023; Pan et al., 2022; Zeng et al., 2025; Zhao et al., 2020). Metaproteomic annotation confirmed that these core MAGs primarily contained functional proteins for hydrolysis and acidogenesis (e.g., KOs encoding proteases and acetate kinase; see Supplementary Table 2.7), which genetically underpin their central role in carbon metabolism (metaproteomic quality control information is provided in Supplementary Figure 3). These results indicate that the primary ecological effect of micro-aeration is to reshape the system's carbon-metabolic network, enriching a group of native fermentative bacteria capable of adapting to micro-oxic fluctuations and sustaining high metabolic fluxes, rather than directly introducing conventional nitrogen-removal specialists.

To identify the principal functional contributors driving efficient ammonia removal, we screened all MAGs harboring nitrogen metabolism genes and conducted a differential analysis of the expression of their associated proteins (21 MAGs in total; Supplementary Figure 4). The results indicated that, compared to the CK group, several key proteins involved in hydroxylamine oxidation and downstream nitrogen oxide reduction were significantly upregulated in the MG group ($p < 0.05$). These proteins included hydroxylamine oxidoreductase (Hao), nitrite reductase (Nir), nitric oxide reductase (Nor), and nitrous oxide reductase (Nos) (Supplementary Figures 5a and 5b). By tracing the detected active proteins back to their host MAGs, we identified that the primary contributors of these nitrogen metabolism functional proteins were the aforementioned taxa enriched under micro-aeration conditions and possessing hydrolytic/acidogenic potential (MAG478, MAG534, MAG434, and MAG208). Collectively, their relative abundance accounted for 44.54%−82.55% of the total abundance of the 21-candidate nitrogen-removing MAGs. Specifically, the Hao protein, responsible for hydroxylamine oxidation, was primarily derived from MAG478 and was highly expressed in the MG group but undetectable in the CK group (Fig. 6). The nitrite-reducing NirK/S and nitrous oxide-reducing NosZ proteins were predominantly expressed in MAG208, MAG534, and MAG478, with their abundance upregulated 1.37- to 24.69-fold compared to the CK group (Fig. 6). Additionally, the metaproteomics data revealed potential denitrification functions of other non-dominant populations. For instance, the Hao protein of *Aminidesulfovibrio aminophilus* (MAG349) and *Methylocaldum* sp. (MAG600) was also upregulated in the MG group. *PFJX01* sp. (MAG415) was found to express the denitrification proteins NirS and NorB/C, which were not detected in the

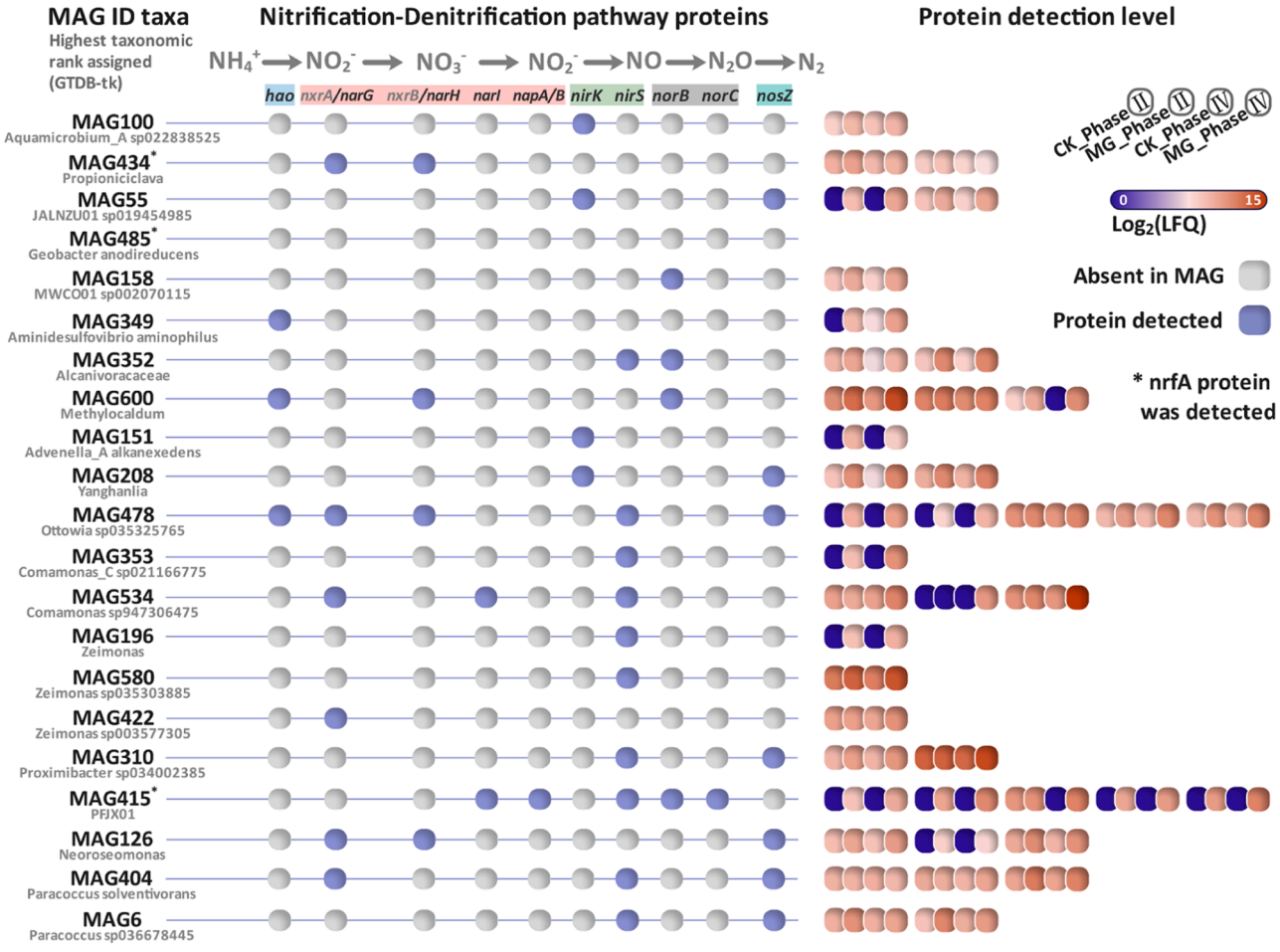


Fig. 6. Proteomic landscape of nitrogen-metabolism proteins in dominant MAGs. The left panel shows the taxonomic assignments of the recovered MAGs determined by GTDB-Tk. The middle panel presents a presence–absence matrix of key proteins involved in nitrogen redox transformation. Solid blue circles indicate detected proteins, whereas gray circles denote absence. The right panel shows the quantitative expression profiles of the detected proteins across treatments. Heatmap color intensity reflects mean protein abundance, expressed as $\log_2$LFQ, from three biologically independent replicates ($n = 3$).

CK group. These non-dominant populations may have collaborated with the core functional taxa to establish a micro-oxygen-induced nitrogen transformation network. Collectively, these findings suggest that, under micro-aeration conditions, nitrogen metabolism is not solely mediated by a single species but rather by a microbial community centered around multifunctional fermentative bacteria, such as MAG478 and MAG534, alongside other populations with nitrogen-metabolism potential, including MAG349, MAG600, and MAG415.

Notably, within the nitrogen transformation functional network described above, the initial activation step from ammonium to hydroxylamine remains unresolved based on the available data. Although low-abundance *amoABC* homologs were detected in the unbinned gene catalog (Supplementary Figure 6), these genes were not assigned to any high-quality MAGs, and no corresponding ammonia monooxygenase (AMO) protein expression was identified in the metaproteome. Regarding downstream nitrogen transformation, no complete nitrite oxidoreductase gene module was identified among the 21 selected candidate MAGs, indicating that the system lacked the genetic potential to further oxidize nitrite to nitrate. Meanwhile, despite the dominant enriched MAG434 carrying partial nitrate-reduction-related genes, the complete respiratory nitrate reductase complex (NarGHI) was not detected in the metaproteome, suggesting that the canonical nitrate reduction pathway was likely not substantially activated under micro-aeration conditions (Fig. 6). Together with the enhanced expression of Hao and the denitrification-related proteins Nir/Nor/Nos described above, these results indicate that ammonium transformation in the micro-aerated system did not conform to a canonical nitrifier-driven nitrification process, but rather represented an atypical nitrogen transformation process involving multiple nitrogen redox functional modules.

3.3. Carbon availability drives metabolic flexibility in nitrogen-removal pathways

After identifying the core functional taxa involved in nitrogen metabolism, we further examined how the system responded to carbon fluctuations inherent to AD. The results revealed distinct metabolic features under carbon-sufficient versus carbon-limited conditions.

Under carbon-sufficient conditions, the system exhibited efficient heterotrophic nitrogen removal. Specifically, in Phase IV, following glucose supplementation, biodegradable COD was rapidly consumed (Fig. 2c), indicating robust heterotrophic metabolic activity. During this phase, MAG208, MAG534, and MAG478 — significantly enriched in the MG group — emerged as the dominant populations (Fig. 5). Metaproteomic data indicated that the Hao protein in MAG478 was specifically expressed, reaching peak abundance during this phase (Fig. 6). Denitrification-related proteins, including NirS in MAG534, NirS and NosZ in MAG478, and NirK and NirS in MAG208, were also enriched to their highest levels (Fig. 6). Furthermore, nitrogen-metabolism proteins from other non-dominant taxa enriched in the MG group (e.g., MAG415, MAG600, MAG349) were similarly upregulated (Fig. 6). The expression profile of all active nitrogen-metabolism proteins during this phase (Fig. 6) aligned with the highest observed nitrogen-removal rate (23.41 $mg{\cdot}L^{-1}{\cdot}d^{-1}$, Fig. 2d) and the rapid conversion of $^{15}NH_4^+$ to $^{15}N_2$ within the first 0–5 days of the isotope-tracing experiment (Fig. 3). Collectively, these findings suggest that, under carbon-sufficient conditions, micro-aeration-enriched versatile fermentative bacteria facilitate efficient atypical nitrogen transformation by upregulating heterotrophic nitrogen-metabolism modules associated with hydroxylamine oxidation and denitrification.

Under carbon-limited conditions, the system sustained its nitrogen removal activity through various endogenous strategies. Specifically, in Phase II, when external carbon was depleted, the nitrogen removal rate remained at 20.07 $mg\ L^{-1}\ D^{-1}$ (Fig. 2d), and isotope tracing confirmed the continued production of $^{15}N_2$ between days 10 and 20 (Fig. 3). Multi-omics analysis revealed that the microbial community employed four major metabolic responses to cope with carbon scarcity and sustain nitrogen metabolism functions.

First, potential hydrogen-utilizing autotrophic metabolism. Hydrogen accumulation was detected in Phase II (Fig. 2b). Concurrently, the abundances of [NiFe]-hydrogenase (HyaAB) and Calvin-cycle carbon-fixation enzymes (RuBisCO and Prk) in MAG415 were found to be 1.51- to 3.23-fold higher than those in the CK group at the same stage (Supplementary Figure 7). This coordinated enhancement suggests that MAG415 may participate in nitrogen removal under carbon-limited conditions through enhanced hydrogen utilization and autotrophic carbon fixation.

Second, intracellular reserve mobilization. In Phase II, metaproteomics analysis revealed that 3-hydroxybutyrate dehydrogenase (Bdh, $log_2LFQ = 7.87$)—which is involved in polyhydroxybutyrate (PHB) degradation—and FadA/B ($log_2LFQ = 6.95$ and 11.67)—which are involved in fatty acid β-oxidation—were specifically induced in *Ottowia* sp. (MAG478) (Fig. 7). These proteins were all detected at near-background levels in the CK group. This expression pattern suggests that, under carbon-limited conditions, MAG478 may enhance the catabolism of intracellular reserves, including PHB and lipids (Fig. 7a), thereby providing endogenous carbon sources and reducing power for cellular maintenance and nitrogen transformation (Wang et al., 2026).

Third, environmental resource scavenging and carbon-conserving metabolism. In Phase II, the branched-chain amino acid transport-related proteins LivG, LivF, and LivK in *Ottowia* sp. (MAG478) exhibited log_2LFQ values of 9.98, 9.51, and 14.85, respectively (Fig. 7), indicating a strong potential for the uptake of residual amino acids. Similarly, in *Comamonas* sp. (MAG534), LivG, LivF, and LivK reached log_2LFQ values of 11.55, 12.11, and 14.68, respectively, while the peptidase PepN attained a log_2LFQ value of 9.85, suggesting an enhanced potential for recovering and utilizing residual amino acids, peptide substrates, and proteinaceous organic nitrogen (Peng et al., 2025; Zhou et al., 2025). Meanwhile, the abundances of multiple proteins associated with the glyoxylate cycle and central carbon metabolism in MAG478 and MAG534 were higher than those in the CK group, including AceA/AceB and Mdh in MAG478 ($log_2LFQ = 10.58$, 9.19, and 9.03, respectively) and AceB and Mdh in MAG534 ($log_2LFQ = 11.87$ and 14.49, respectively) (Fig. 7). These findings suggest that under carbon-limited conditions, MAG478 and MAG534 may enhance the acquisition efficiency of limited substrates and redistribute carbon flux through the glyoxylate shunt and TCA-related metabolism to maintain the supply of precursors and reducing power (Peng et al., 2025).

Fourth, alternative substrate utilization and respiratory adaptation. In Phase II, MAG478 exhibited enhanced expression of proteins associated with respiratory electron transfer, with Cyt c, PetA, and NuoD/G specifically detected, while the abundances of PetC and AtpA increased by 347.6% and 169.0%, respectively, compared to the CK group. Terminal respiration-related proteins, including CoxA/B, were also detected (Fig. 7b). Together with the upregulation of Hao, these observations suggest a potential link between Hao-mediated NH_2OH oxidation and cytochrome-mediated electron transfer and energy conservation. Meanwhile, *Methylocaldum* sp. (MAG600) upregulated proteins involved in the oxidation of methanol to CO_2, indicating its potential to utilize C1 compounds as alternative carbon sources (Supplementary Figure 8). Additionally, this strain, along with *Yanghanlia* sp. (MAG208), demonstrated elevated expression levels of high-affinity cytochrome oxidases (e.g., CydA and CcoN/O) (Fig. 7 and Supplementary Figure 9), suggesting that some functional taxa may sustain energy metabolism under low-oxygen and low-carbon conditions through alternative substrate utilization and respiratory chain adaptation (Inskeep et al., 2025; Steimle et al., 2021).

3.4. Genetic basis of the micro-aeration-driven "Methane-for-nitrogen" trade-off

The apparent methane–nitrogen trade-off coefficient quantitatively

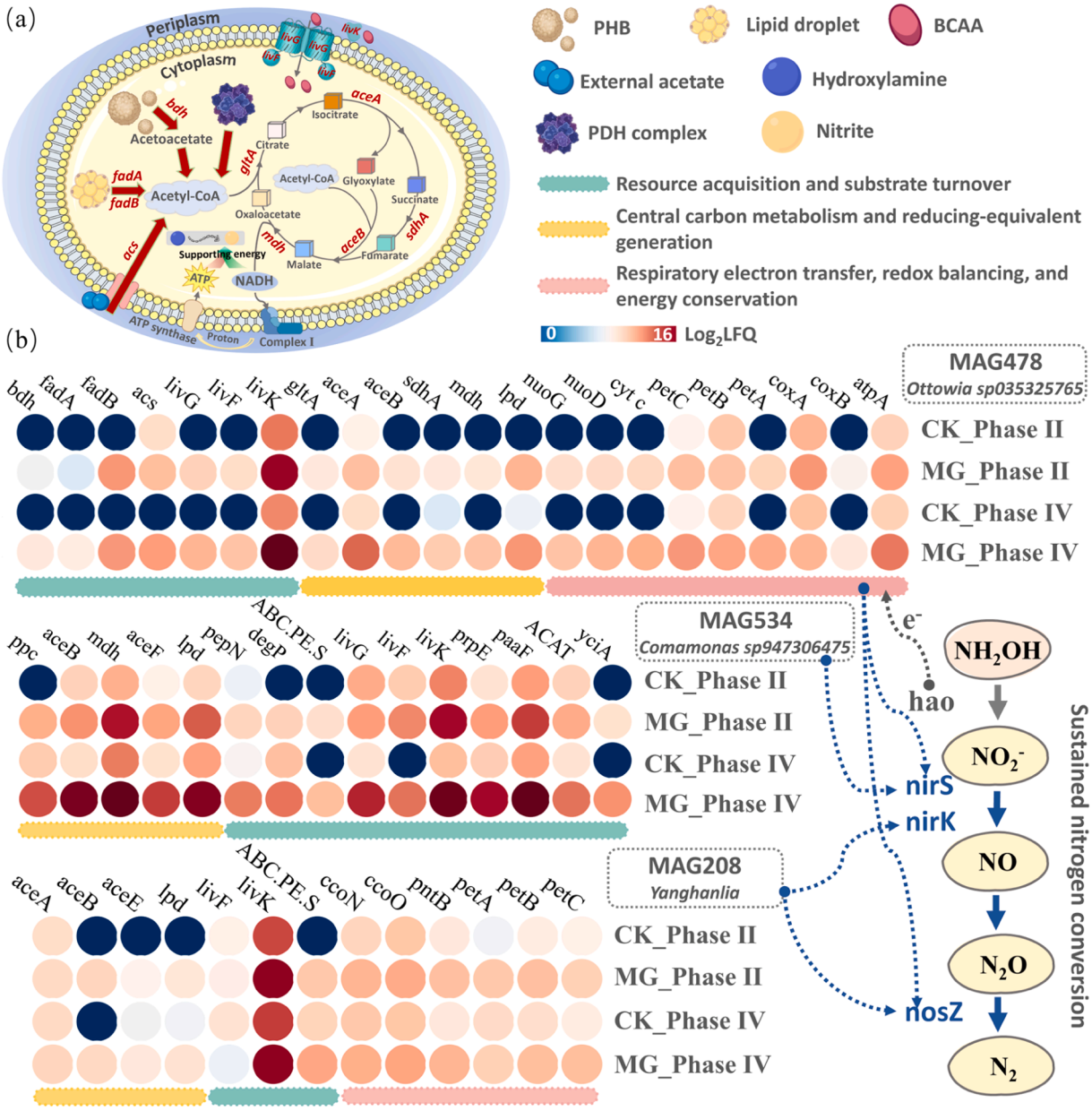


Fig. 7. Metabolic and respiratory strategies of three dominant MAGs sustaining nitrogen transformation under carbon-limited conditions. (a) Potential links among hydroxylamine oxidation, carbon-metabolic adaptation, and respiratory energy conservation in MAG478 under carbon-limited conditions. Hao catalyzes NH_2OH oxidation and may be coupled to cytochrome-mediated respiratory electron transfer and energy conservation. (b) Key proteins detected in MAG478, MAG534, and MAG208, grouped into resource acquisition and substrate turnover, central carbon metabolism and reducing-equivalent generation, and respiratory electron transfer, redox homeostasis, and energy conservation. Dashed lines indicate the MAG-level assignment of the corresponding nitrogen-transformation functions; gray dashed arrows indicate putative electron-transfer links inferred from metagenomic and metaproteomic evidence.

captures the balance between the activation of nitrogen removal and the suppression of methanogenesis under micro-aeration conditions. To further elucidate the genetic basis underlying this trade-off, we analyzed changes in the abundance of functional genes associated with methanogenesis (Supplementary Figure 10). The metagenomic data revealed that, in the MG group, genes associated with methyl-coenzyme M reductase, formylmethanofuran dehydrogenase, and tetrahydromethanopterin S-methyltransferase decreased by 34.32%–69.37%, 43.66%–70.20%, and 50.22%–74.89%, respectively, compared with the CK group, and functions associated with acetoclastic methanogenesis decreased by 10.62%–24.61% (Supplementary Figure 10). These changes indicate that micro-aeration conditions weakened the functional potential for methanogenesis (Peng et al., 2023). Meanwhile, ROS-detoxification functions, including superoxide dismutase (SOD), catalase, and peroxiredoxin, were markedly restructured, with their combined abundance decreasing by 19.89% and 45.77% in Phases II and IV, respectively (Supplementary Table 2.8).

4. Discussion

4.1. Pulsed micro-aeration creates transient ecological windows for nitrogen removal in AD

Conventional biological nitrogen removal in wastewater treatment relies on the establishment of stable aerobic and anoxic environments. In this study, low-frequency pulsed aeration, applied once daily, was used to generate transient and non-steady-state micro-aeration events within the reactor. During and shortly after each aeration pulse, local dissolved

oxygen and ORP transiently increase, forming a dynamic and rapidly decaying micro-oxic gradient (Fig. 4b). Although this gradient persists only briefly and returns to anaerobic levels within minutes, it creates periodically occurring functional windows for facultative or oxygen-tolerant microorganisms, such as *Ottowia* and *Comamonas* (Nguyen et al., 2019). These windows provide transient electron acceptors for ammonium conversion, while the subsequent anaerobic period facilitates denitrification. Our findings indicate that precise regulation of the intensity and frequency of pulsed aeration can activate biological nitrogen removal within AD systems at relatively low oxygen input without disrupting the bulk anaerobic environment of the AD reactor. To the best of our knowledge, previous studies have not reported the induction of such a nitrogen transformation process in AD systems through pulsed micro-aeration.

4.2. Versatile fermentative bacteria drive cross-boundary nitrogen metabolism

One of the most striking findings of this study is that the core microorganisms driving the nitrogen-removal process were not traditional nitrifiers (e.g., ammonia-oxidizing bacteria and nitrite-oxidizing bacteria) or dedicated denitrifiers, but rather typical hydrolytic-acidogenic bacteria (Fig. 5). Consistent evidence from metagenomics and metaproteomics demonstrates that these bacteria, under micro-aeration stimulation, upregulate functional modules associated with hydroxylamine oxidation (Hao) and the complete denitrification cascade (Nir, Nor, and Nos) (Fig. 6). This finding sharply contrasts with the highly specialized functional guilds commonly employed in wastewater treatment, such as *Nitrosomonas, Nitrospira, Nitrobacter*, and *Paracoccus* (Mai et al., 2021; Su et al., 2023). It suggests an important ecological mechanism: in response to the environmental perturbation of micro-aeration, metabolically versatile generalists in the AD system may possess a competitive advantage over specialized experts. These hydrolytic-acidogenic bacteria inherently carry the genetic information for complete nitrogen metabolism modules. Stimulated by micro-aeration conditions, they shift their metabolic focus from conventional hydrolysis-acidogenesis towards nitrogen transformation by upregulating the expression of these genes. This enables them to occupy a new niche at the interface of carbon and nitrogen cycles, thereby playing a critical role as cross-boundary functional executors. This finding expands our understanding of functional redundancy and metabolic-network complexity in complex anaerobic communities.

However, the initial activation step from ammonium to hydroxylamine has not been directly resolved in this study. Previous research has indicated that certain heterotrophic nitrification-aerobic denitrification bacteria can perform ammonia oxidation or produce hydroxylamine even in the absence of recognizable amoA/AMO homologs (Qin et al., 2024; Silva et al., 2020), thereby suggesting the presence of non-canonical ammonium activation mechanisms. Consequently, both the low-abundance *amoABC*-associated genetic potential and the yet-to-be-resolved non-canonical ammonium activation mechanisms may play a role in the initial conversion of ammonium to hydroxylamine. However, the specific contributions and molecular mechanisms of these processes require further investigation.

4.3. Endogenous metabolic flexibility distinguishes this process from conventional processes

The micro-aeration-driven nitrogen removal process demonstrated in this study exhibits remarkable metabolic adaptability, characterized by its ability to sustain efficient nitrogen removal under both carbon-sufficient and carbon-limited conditions. Conventional wastewater treatment systems typically operate under carbon-sufficient conditions but generally require the addition of organic carbon sources (e.g., methanol and acetate) to maintain denitrification when the influent carbon-to-nitrogen ratio is low (Pan et al., 2023). In contrast, when carbon was depleted in our system, the nitrogen removal flux was sustained through a dual strategy: (i) the potential utilization of endogenously produced hydrogen for hydrogen-utilizing autotrophic metabolism via the enriched MAG415 (Supplementary Figure 7), and (ii) the activation of endogenous reserve mobilization and environmental necromass recycling by core functional bacteria (MAG534, MAG478, and MAG208) to sustain nitrogen removal (Fig. 7).

Together, these findings reveal a deeper mechanism: the introduction of micro-aeration conditions does not merely facilitate nitrogen removal but actively stimulates the inherent endogenous electron-donor potential of the AD system, which typically remains underutilized in strictly anaerobic environments. In a conventional strictly anaerobic state, hydrolysis and acidogenesis products, along with microbial intracellular reserves, are primarily directed towards methanogenic pathways. However, micro-aeration intervention reshapes this carbon flow allocation by redirecting electron donors towards nitrogen removal pathways. This enables the system to sustain efficient nitrogen removal even under carbon-limited conditions. This metabolic transition, from reliance on exogenous electron donors to the utilization of endogenous electron donors, reflects a distinctive survival strategy of anaerobic digestion microbiomes under micro-aeration conditions. The efficient nitrogen removal observed in real digestate further substantiates the activation of the endogenous electron-donor potential of this system through micro-aeration.

4.4. "Methane-for-nitrogen" trade-off prompts a spatially decoupled process concept

The activation of nitrogen removal under micro-aeration was accompanied by suppressed methane production, indicating a pronounced metabolic trade-off between the two processes. This inhibition may be driven by multiple mechanisms. First, micro-aeration introduces oxidative stress, as low-level O_2 exposure can directly inhibit methanogenic activity. Morais et al. (2024) reported that an O_2 concentration of only 0.5% reduced initial methane production rates from H_2/CO_2 and acetate by approximately 31% and 39%, respectively. Additionally, the increase in ORP induced by micro-aeration further disrupts the strongly reducing environment that methanogens require, potentially inducing pronounced oxidative-stress responses and proteomic remodeling (Kleikamp et al., 2024). In the present study, the observed decrease in the combined abundance of genes associated with ROS detoxification, including SOD, catalase, and peroxiredoxin (Supplementary Table 2.8), provides indirect evidence that micro-aeration disrupted the oxidative-stress defense system of the microbial community. Second, the consumption of organic carbon and reducing equivalents during nitrogen removal can divert substrates and electrons that would otherwise be available for methane production, redirecting carbon and electron flows from CH_4 generation toward nitrogen transformation. Recent studies have also indicated that enhanced denitrification is often accompanied by a reprogramming of carbon metabolism and reducing-power supply, thereby altering electron allocation within the system (Peng et al., 2025). Collectively, these two mechanisms may jointly contribute to the trade-off between nitrogen removal and methanogenesis under micro-aeration conditions.

Notably, the apparent methane–nitrogen trade-off coefficient exhibited a decreasing trend throughout the operational phases, indicating that the system may undergo a degree of adaptive optimization under sustained micro-aeration conditions. However, even if long-term acclimation may partially alleviate methane losses, the intrinsic competition for electron donors between nitrogen removal and methanogenesis is unlikely to be entirely resolved within a single reactor. Therefore, spatial decoupling of these two processes may offer a more reliable solution to the "methane-for-nitrogen" dilemma.

Based on these insights, we propose a spatially decoupled process configuration: "mainstream anaerobic methanogenesis–side-stream micro-aerobic nitrogen removal" (Fig. 8). This configuration spatially

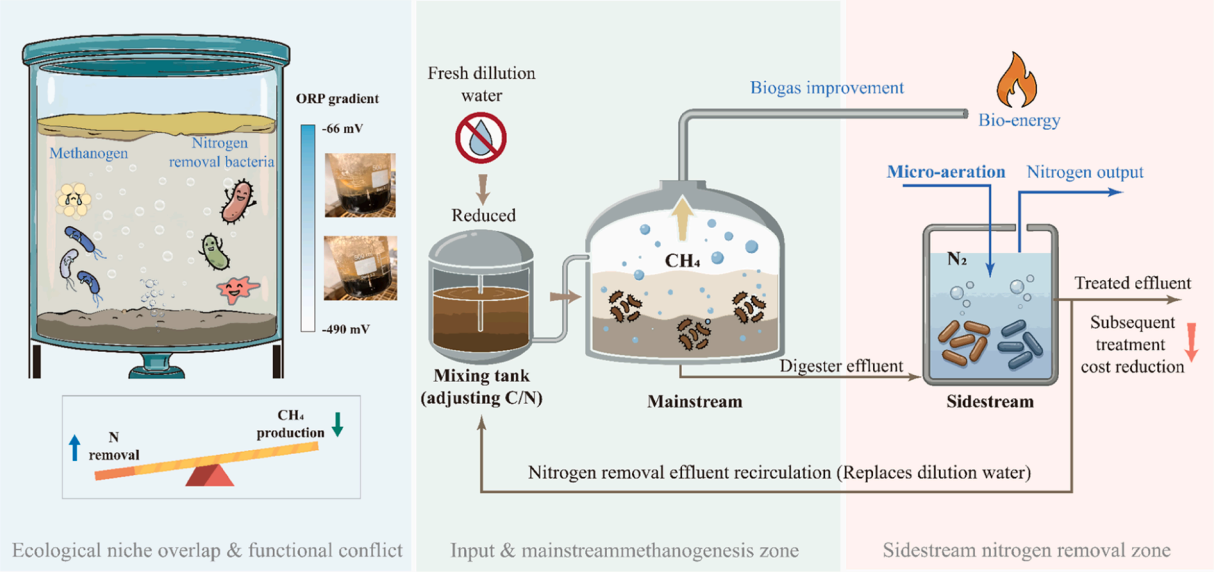


Fig. 8. Spatially decoupled "mainstream methanogenesis and side-stream micro-aerobic nitrogen removal" paradigm for overcoming the methane-for-nitrogen tradeoff in AD. The design preserves the oxygen-sensitive mainstream methanogenic process under strict anaerobic conditions, thereby maximizing energy recovery. An ammonia-rich digestate side-stream is redirected to a micro-aerobic unit for efficient nitrogen removal. The treated low-ammonia effluent is then recirculated to dilute the mainstream, regulate the feed C/N ratio, facilitate internal water recycling, and decrease the demand for fresh dilution water.

separates the two competing processes: the mainstream reactor is maintained under strictly anaerobic conditions to maximize methane production from organic matter for energy recovery, while ammonia-rich digestate is diverted to a side-stream micro-aerobic reactor for nitrogen removal. Spatial decoupling also facilitates the separation of N_2 from biogas, thereby preventing the negative impact of N_2 on the calorific value and purity of biogas. The low-ammonia effluent from the side-stream unit can be valorized by substituting most, or even all, of the fresh water used for feedstock slurry preparation, without causing ammonia inhibition in the AD reactor. The functional microorganisms present in the recirculated stream may further enhance the robustness of the system.

The core innovation of this process resides in the side-stream micro-aerobic unit itself. Unlike conventional biological nitrogen removal processes, this unit requires only low-dose pulsed aeration to trigger endogenous-sustained nitrogen removal, thereby eliminating the need for high dissolved oxygen levels and external carbon sources typically required for conventional nitrification. This advancement has the potential to substantially reduce both energy and chemical costs. Even when operated solely as a digestate treatment unit without recirculation, this configuration can achieve significant cost savings. Nevertheless, several uncertainties remain regarding the scale-up of this unit. Among these uncertainties, maintaining spatial heterogeneity within the micro-aerobic reactor poses a significant challenge, as variations in aeration modes, reactor geometries, and hydrodynamic characteristics can influence the spatiotemporal distribution of oxygen. This, in turn, may constrain the enrichment efficiency of functional microorganisms and the stability of nitrogen removal. Additionally, future studies should systematically evaluate potential issues related to side-stream micro-aerobic nitrogen conversion, including N_2O emissions (Flores-Alsina et al., 2025), the impact of EPS accumulation on sludge dewaterability (He et al., 2024), and the risk of residual dissolved oxygen inhibiting mainstream methanogenesis (Morais et al., 2024). Supplementary Table 2.9 provides a preliminary engineering boundary comparison between this technology and conventional ammonia inhibition mitigation strategies (e.g., dilution, co-digestion, and chemical stripping); however, a more systematic quantitative comparison is contingent upon long-term operational data from continuous-flow scaled-up systems.

5. Conclusion

Pulsed micro-aeration effectively activated the conversion of ammonia nitrogen to N_2 within an AD system, without compromising the bulk anaerobic environment. ^{15}N isotope tracing confirmed that approximately 82.49% of the transformed labeled $^{15}NH_4^+$-N was recovered as $^{15}N_2$, establishing N_2 as the primary terminal product of nitrogen transformation. The key functional executors are not traditional nitrifiers or denitrifiers, but rather native fermentative/facultative heterotrophic bacteria present in the AD system, notably *Ottowia* and *Comamonas*, whose functional remodeling was closely associated with this process. Under micro-aeration conditions, these populations upregulated proteins associated with hydroxylamine oxidation, nitrogen oxide reduction, and respiratory electron transfer, demonstrating distinctive metabolic flexibility: under carbon-sufficient conditions, heterotrophic denitrification predominates; conversely, under carbon-limited conditions, the system shifts toward endogenous-sustained modes involving potential hydrogen utilization, mobilization of intracellular reserves (PHB and fatty acids), residual organic matter scavenging, and the glyoxylate cycle. Notably, canonical ammonia monooxygenase proteins were not detected, and the initial ammonium activation step remains unresolved. However, the activation of nitrogen removal by micro-aeration was accompanied by suppression of methanogenesis, with a methane loss of 1.31–2.77 mL per mg of TAN removed. Based on this understanding, we propose a spatially decoupled configuration integrating "mainstream anaerobic methanogenesis–side-stream micro-aerobic nitrogen removal." Future studies should systematically validate the long-term operational stability of this configuration in continuous-flow reactors, optimize the side-stream ratio, and control oxygen carryover to advance its practical implementation.

CRediT authorship contribution statement

Linpei Han: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Lei Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Wenjie Ye:** Visualization, Methodology. **Jiaqi Huang:** Visualization, Methodology. **Yongli Liu:** Visualization, Conceptualization. **Chaoran Duan:** Visualization, Methodology. **Bin Zhan:** Visualization, Methodology. **Shangbang Guo:**

Visualization, Methodology. **Xuya Peng:** Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No 52470139), the Natural Science Foundation of Chongqing Municipality, China (Grant No CSTB2024NSCQ-QCXMX0088).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2026.126838.

Data availability

The authors do not have permission to share data.

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